Introduction

In this assignment we are testing a pre-trained YOLO model with a trained YOLO model. The dataset used for this assignment is the LISA road signs dataset that came from Roboflow. Here is a link to that dataset: https://universe.roboflow.com/dakota-smith/lisa-road-signs. The dataset came with annotation files so no extra annotations were made. Overall, the trained model proved to be a success with detection on static images, but struggled a bit with frame by frame detection in videos.

Dataset and Preprocessing

Like what was said in the introduction, the LISA road signs dataset was used for this YOLO model. It contains roughly six and a half thousand images split amongst testing, validation, and training images. Accompanying those images are annotation files that highlight specific signs that need to be identified. This dataset likes to split each image into a particular sign to look for. It is about a 50/50 split for images in color and black and white with most images being in between 480p and 522p.

Very little preprocessing was done with the dataset as the dataset provided already had preprocessing by default. The dataset also had a lot of repeat images from different snapshots in time so the small signs were at different locations better improving the accuracy of the model. No data augmentation was used within the dataset either.

Challenges

Many challenges were faced trying to detect small objects within photos; however, a lot of challenges came from videos. The biggest challenge faced throughout this project was trying to get the model to detect signs correctly. It can detect fairly small signs fairly easily, but accuracy unfortunately suffered. One of the examples was the no left/right turn sign being

confused with a stop sign. The problem looks to be the color of the sign interfering with the model which is surprising as the training data did include a lot of non-color images to help correlate shapes to the signs as opposed to just looking at the pixel colors. Perhaps the training data included more color images for the no left/right turn and stop signs than other signs leading to worse overall results.

Another challenge faced during this project was dealing with false positives. There were a few examples when the model would give a false positive on random objects. Weirdly, the only false positive seemed to be pedestrian crossing signs. This probably happened because the pedestrian sign is a bit harder to detect and train due to its internal shape. Overall, the model did pretty well all things considered and was able to detect many small signs in the dataset.

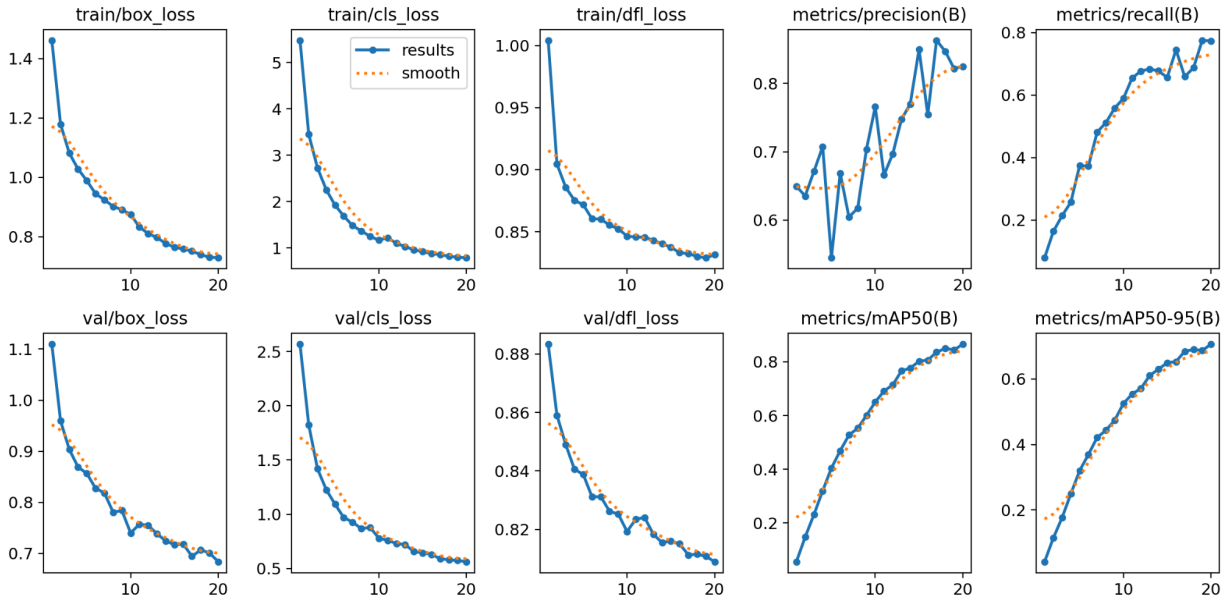
Results

All results except for the video were predicted using an image size of 640 as the machine used to test and train could not reliably complete testing and training with the amount of RAM needed. In terms of confidence the trained model was way better than the pre-trained model at detecting objects within the dataset. Many times the pre-trained model would mistake certain elements of the scene to be different things or just not detect them altogether.

The trained model was trained at 640 image size with 16 batches and 20 epochs. It was able to discern signs very well with confidence levels never falling below that 50 percent threshold with most detections landing a commendable >85 percent confidence level. When comparing the pre-trained model many times a lower confidence level had to be put into place just for the model to detect certain objects even though at a roughly 30-50 percent accuracy. However, the trained model did struggle with confusing signs when given a video and would accidentally label some signs incorrectly.

Suggestions

The small object detection in this assignment would have had a bigger impact if the model could have been trained using 30 epochs and 1280 image size; however, the machine used to train the model just couldn't handle the dataset with training. The model itself didn't really struggle too much though with its lowered image size and epochs. Looking at the graphs provided by the trainer it seemed like it was already hitting diminishing returns and leading into the category of overfitting with just 20 epochs. One thing that it could have helped with was detection on the video portion as it did really struggle there.



Overall, it is very fascinating to see models detect very small objects that are sometimes just a few pixels in size. At that size it would be hard for anyone to see those objects. With better training and potentially even bigger datasets the image detection would be incredible and potentially be able to beat a trained human eye at some examples.